

Calc 1 – Continuity

Worksheet A

1. Classify the discontinuity of $f(x) = \frac{x^2 - 1}{x - 1}$ at $x = 1$.
2. Find the value of c that makes $f(x) = \begin{cases} x^2 + 1, & x \leq 2 \\ cx + 3, & x > 2 \end{cases}$ continuous on $\mathbb{R}$.
3. State where $f(x) = \frac{1}{x^2 - 4}$ is continuous; classify each discontinuity.
4. Use IVT to show $x^3 + x - 1 = 0$ has a root in $(0, 1)$.
5. Does $f(x) = \lfloor x \rfloor$ satisfy the conclusion of the EVT on $[0, 2]$? Explain.

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Worksheet B

1. Classify the discontinuity of $f(x) = \frac{|x|}{x}$ at $x = 0$.
2. Find a, b such that $f(x) = \begin{cases} ax + 1, & x < 1 \\ 4, & x = 1 \\ x^2 + b, & x > 1 \end{cases}$ is continuous at $x = 1$.
3. Determine all points of discontinuity of $f(x) = \tan x$ on $[-2\pi, 2\pi]$.
4. Show $\cos x = x$ has a solution in $(0, 1)$.
5. Give an example of a function on $(0, 1)$ that is continuous but has no maximum. Why doesn't this contradict EVT?

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Answer Key (Worksheets A & B)

Worksheet A.

1. $f(x) = x + 1$ for $x \neq 1$; $f(1)$ undefined, $\lim = 2$. **Removable.**
2. Continuity at 2: $2^2 + 1 = 2c + 3 \Rightarrow 5 = 2c + 3 \Rightarrow \boxed{c = 1}$.
3. Continuous on $\mathbb{R} \setminus \{\pm 2\}$. At $x = \pm 2$ the denominator $\rightarrow 0$ with $f \rightarrow \pm\infty$, so **infinite discontinuities** (vertical asymptotes).
4. $f(x) = x^3 + x - 1$ is a polynomial (continuous). $f(0) = -1 < 0$, $f(1) = 1 > 0$. By IVT $\exists c \in (0, 1)$ with $f(c) = 0$.
5. $\lfloor x \rfloor$ is *not* continuous on $[0, 2]$ (jumps at 1 and 2), so EVT does not apply. Nonetheless, it does attain a max (2) and a min (0) on $[0, 2]$ — EVT is sufficient, not necessary.

Worksheet B.

1. Left limit $= -1$, right limit $= +1$. **Jump discontinuity.**
2. Need $a + 1 = 4$ and $1 + b = 4$. So $\boxed{a = 3, b = 3}$.
3. $\tan x$ undefined at $x = \pi/2 + k\pi$. On $[-2\pi, 2\pi]$: $x = \pm\pi/2, \pm3\pi/2$ — four **infinite discontinuities**.
4. Let $g(x) = \cos x - x$. Then $g(0) = 1 > 0$ and $g(1) = \cos 1 - 1 < 0$. g is continuous, so by IVT $\exists c \in (0, 1)$ with $g(c) = 0$.
5. $f(x) = x$ on $(0, 1)$ is continuous, bounded above by 1, but never attains 1. No contradiction: EVT requires a *closed* bounded interval; $(0, 1)$ is open.